

Offline Reinforcement Learning for a Small Language Model in Reference Games. A Reproduction Study Using clembench

Imge Yüzüncüoglu
Supervisor: Prof. Dr. Sherzod Hakimov



[Link to GitHub Repository](#)

Motivation

- ❖ RL is increasingly used to fine-tune LLMs for **better generalisation** compared to SFT.
- ❖ Gül & Artzi (2024) reported significant improvements via **joint inference** (generation + comprehension) in a **ReferenceGame (image based)** setup.
- ❖ Reproduction and analyses of those results using open-source tools to test **reproducibility** and **human influence** in quality of fine-tuning LLMs.

Research Questions

- Can the improvements reported by Gül & Artzi be reproduced?
- What is the relative contribution of **human interaction** vs **algorithmic optimisation**?

Experimental Setup

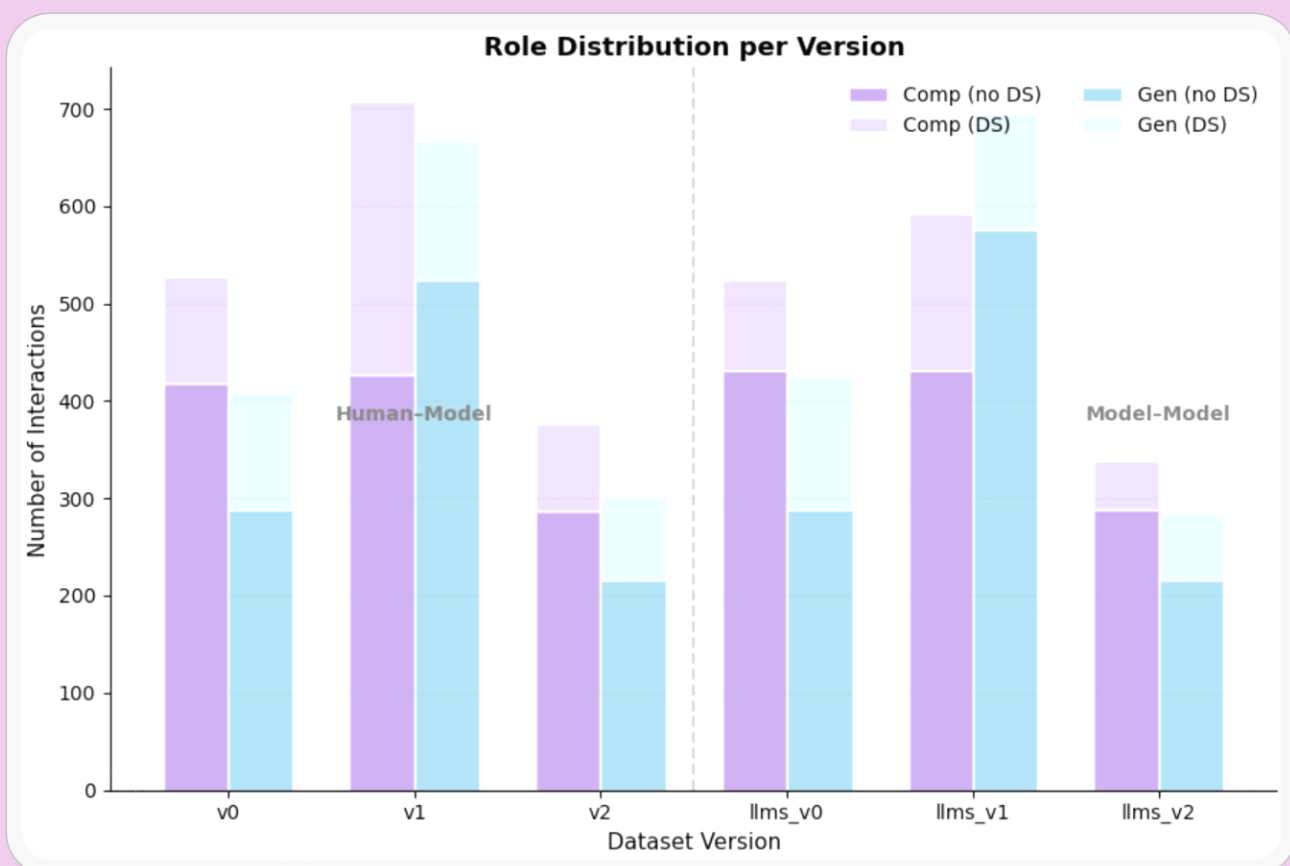
- **The ReferenceGame (Dialogue based)**
 - 2-player, single-turn, collaborative task: **Describer & Guesser**
 - 5×5 grids (letters, numbers, shapes, symbols) e.g.:

x x x x x x x x x x x x x x x x x x x x x □ □ □ x x x x x x
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- Evaluation of comprehension and generation skills.

➤ Data Collection

- Human & Model: 14 participants via Discord/Zoom
- Model & Model: Automated self-play (Baseline)
- Data Sharing
- Data generation → Offline RL fine-tuning → New inference data → Iterative update (v0 → v1 → v2)



➤ Win Ratio:

Model	v0	v1	v2	llms_0	llms_1	llms_2
Gen	0.56	0.63	0.58	0.54	0.4	0.42
Comp	0.44	0.6	0.46	0.44	0.47	0.35

Comment: Base model was Llama3.1-8B-Instruct.

Methodology

- ❖ Objective: maximise reward $r \in \{-1, +1\}$ for correct guesses
- ❖ REINFORCE loss with IPS:
$$L_{REINFORCE+IPS} = -c_l r \log P_\theta(\hat{t} | I, u)$$
- ❖ Early Stopping based on comprehension accuracy
- ❖ Usage of QLoRA and HF's Transformers library

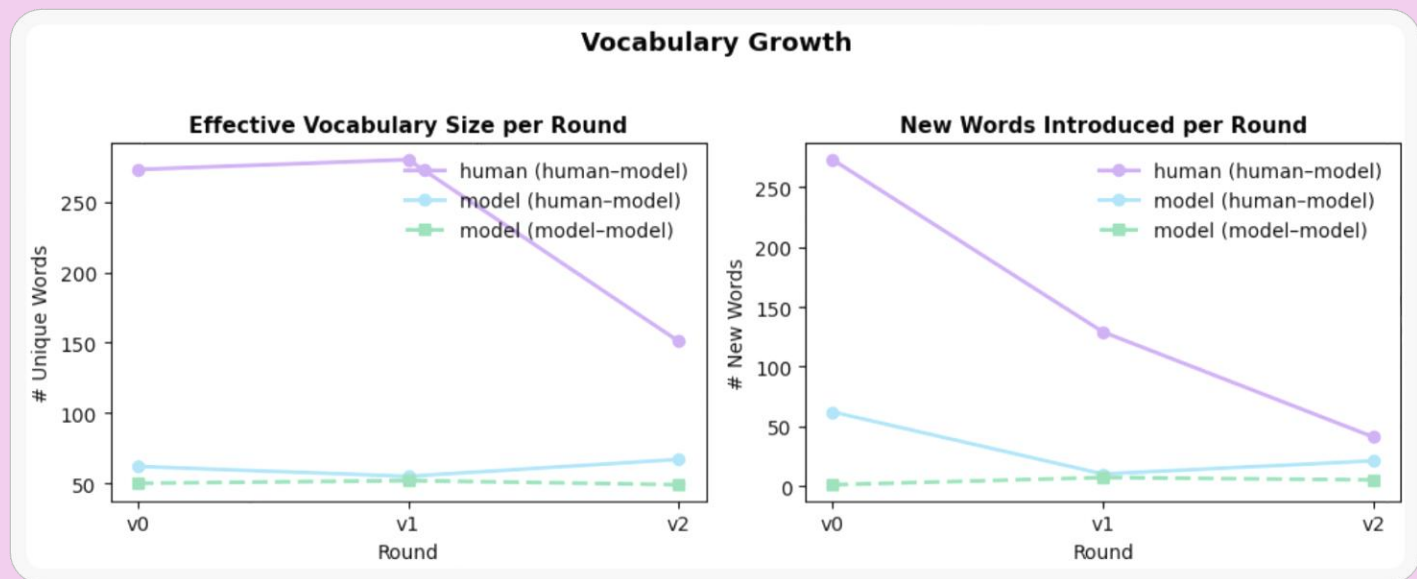
Results

- **Findings from Clembench benchmark v3.0:**
 - RL fine-tuning did not replicate performance boost observed by Gül & Artzi.
 - Human data improved linguistic richness but not task accuracy.
 - Performance dropped in later versions, probably due to data imbalance.

Model	ClemScore (Overall)	ClemScore (ReferenceGame)
v0	31.5	43.3
v1	29.5	40
v2	24.3	36.7
llms_v1	28.9	40
llms_v2	28.6	33.3

➤ Language Analysis (Human & Model):

- *Vocabulary diversity* increased with human input, stabilising over time
- *Utterance lengths* converged between humans and models (~17–24 tokens).
- Common issues:
 - *Miscounting* elements
 - Vague or ambiguous phrasing
 - Lack of contrastive descriptions
- Human input improved *fluency*, but **not** precision or grounding.



GM (Game Master) to Player 1 (InstructionGiver)

You are given three grids, where each of them is 5 by 5 in size.
Grids have empty cells marked with "□" and filled cells marked with "X".
Your task is to generate a referring expression that best describes the target grid while distinguishing it from the two other distractor grids.
The first grid is the target grid, and the following two grids are the distractors.

Target grid:

```
X □ X □ □  
X □ □ X  
X □ □ X  
□ □ □ X  
X □ X X X
```

Distractor grid 1:

```
X □ X X X  
X X X □ X  
X □ □ X X  
X X X □ X  
X □ X X X
```

Distractor grid 2:

```
X X X X X  
X X X □ X  
□ □ □ X X  
□ □ □ X  
X □ X X X
```

Instruction: Describe the target grid.
Generate the referring expression starting with the tag "Expression: " for the given target grid. Omit any other text.

Description from human-model v0

Player 1 (InstructionGiver) to GM (Game Master)

Expression: The grid that has exactly one row with three X's and two rows with two X's.

Description from human-model v1

Player 1 (InstructionGiver) to GM (Game Master)

Expression: The grid that has a row with three X's in the top left, a row with two X's in the middle, and a row with three X's in the bottom right.

Description from human-model v2

Player 1 (InstructionGiver) to GM (Game Master)

Expression: The grid that has exactly one cell in the top-left corner, has a cell in the top-right corner, has two cells in the bottom-left corner, and has four cells in the bottom-right corner.

➤ Limitations

- Limited human participations and manual setup.
- GPU constraints prevented v3.
- Diverse toolchains not optimised for RL.

➤ Future Work

- Browser-based Slurk interface (already done).
- Introduce generation-based early stopping criteria.
- Explore data quality and balance effects.

➤ Conclusion

- Human input influences *how* models speak, but also *how well* they perform?
- RL for small models is promising.
- Reproducibility in RL-style pipelines was the core challenge due to hinderances by fast-evolving dependencies.

➤ Acknowledgements

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➤ References

- Kranti Chalamalasetti, Jana Götze, Sherzod Hakimov, Brielen Madureira, Philipp Sadler, and David Schlangen. 2023. Clembench: Using game play to evaluate chat-optimized language models as conversational agents. Preprint, arXiv:2305.13455.
- Mustafa Omer Gul and Yoav Artzi. 2024. Cogen: Learning from feedback with coupled comprehension and generation. arXiv preprint arXiv:2408.15992.